



LOVELY
PROFESSIONAL
UNIVERSITY

INT217

PROJECT REPORT

(Project Semester January–April 2025)

Crude oil and petroleum products import and export 2023-24

Submitted by:

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Programme and Section: B. Tech & CSE K23GW

Course Code: INT217

Under the Guidance of

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Discipline of CSE/IT

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CERTIFICATE

This is to certify that Pushkaraditya Singh bearing Registration no. 12304124 has completed INT217 project titled, “***Crude oil and petroleum products import and export 2023-24*** ” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Name of the Supervisor: Baljinder Kaur

Lovely Professional University

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Date: 12-04-2025

DECLARATION

I Pushkaraditya Singh student of LPU under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-09-2025

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Signature

Pushkaraditya Singh

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to those who supported me throughout the completion of this project titled “Crude Oil and Petroleum Products Import and Export 2023–24”. First and foremost, I would like to thank Baljinder Kaur

my respected project supervisor, for her invaluable guidance, support, and encouragement. Her expert advice and timely feedback helped me to stay focused and complete this project efficiently.

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A special thanks to my peers, friends, and family for their constant moral support and encouragement throughout the project.

Lastly, I would like to acknowledge the use of the dataset which served as the foundation for the analysis conducted in this report.

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Introduction

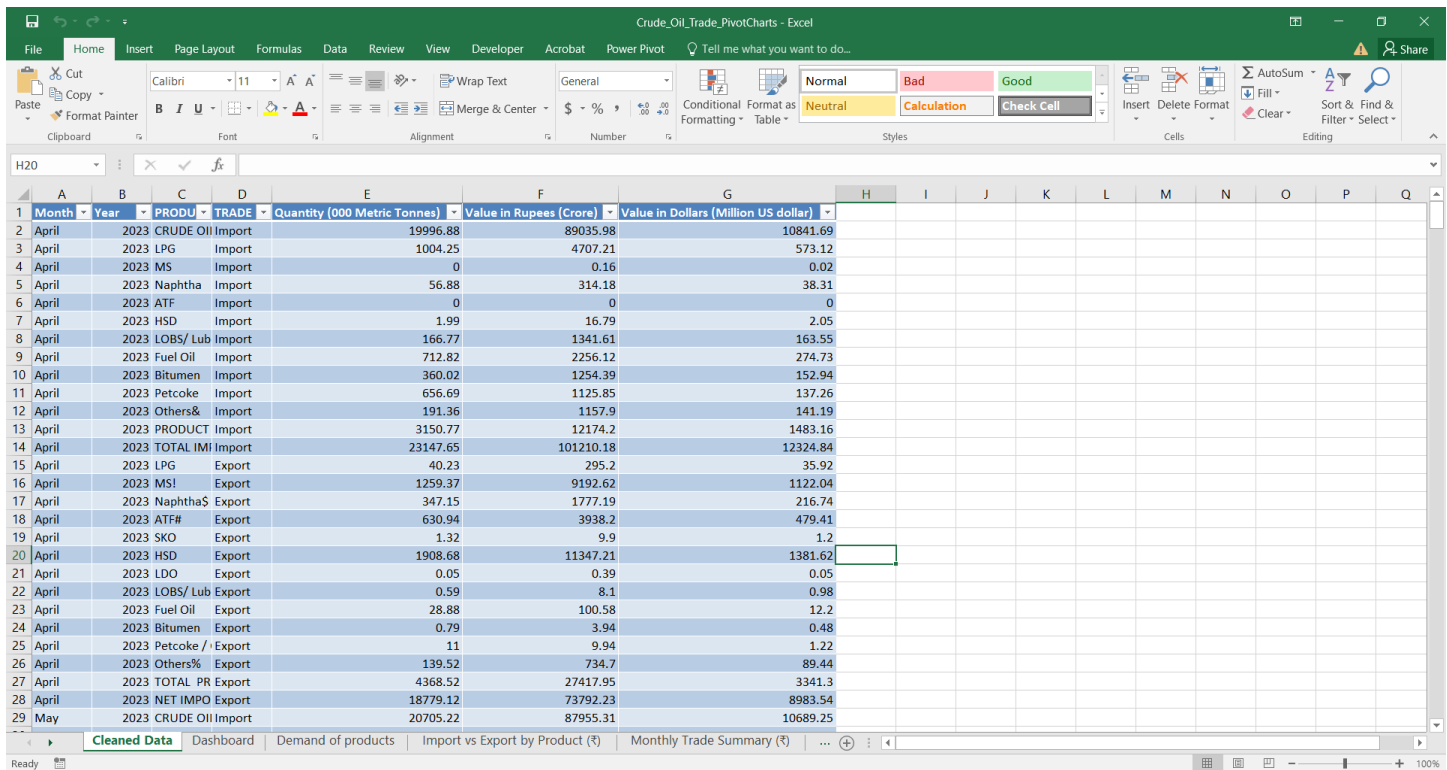
This report presents an in-depth analysis of crude oil and petroleum product imports and exports. It is aimed at understanding trade patterns, key products, country-wise dependencies, and volume trends over time. The report is structured to guide the reader from the raw dataset to insightful visualizations and analytical conclusions.

1. Source of Dataset

The dataset used in this report is derived from official trade records and compiled in an Excel spreadsheet titled **Crude_Oil_Trade_PivotCharts.xlsx**. It contains monthly and yearly trade quantities of crude oil and various petroleum products across different countries. The source is <https://www.data.gov.in/catalog/import-export-petroleum-products>.

2. Sample Dataset

Dataset:



Month	Year	PRODUCT	TRADE	Quantity (000 Metric Tonnes)	Value in Rupees (Crore)	Value in Dollars (Million US dollar)
April	2023	CRUDE OIL	Import	19996.88	89035.98	10841.69
April	2023	LPG	Import	1004.25	4707.21	573.12
April	2023	MS	Import	0	0.16	0.02
April	2023	Naphtha	Import	56.88	314.18	38.31
April	2023	ATF	Import	0	0	0
April	2023	HSD	Import	1.99	16.79	2.05
April	2023	LOBS/ Lub	Import	166.77	1341.61	163.55
April	2023	Fuel Oil	Import	712.82	2256.12	274.73
April	2023	Bitumen	Import	360.02	1254.39	152.94
April	2023	Petcoke	Import	656.69	1125.85	137.26
April	2023	Others&	Import	191.36	1157.9	141.19
April	2023	PRODUCT	Import	3150.77	12174.2	1483.16
April	2023	TOTAL IMI	Import	23147.65	101210.18	12324.84
April	2023	LPG	Export	40.23	295.2	35.92
April	2023	MSI	Export	1259.37	9192.62	1122.04
April	2023	Naphtha\$	Export	347.15	1777.19	216.74
April	2023	ATF#	Export	630.94	3938.2	479.41
April	2023	SKO	Export	1.32	9.9	1.2
April	2023	HSD	Export	1908.68	11347.21	1381.62
April	2023	LDO	Export	0.05	0.39	0.05
April	2023	LOBS/ Lub	Export	0.59	8.1	0.98
April	2023	Fuel Oil	Export	28.88	100.58	12.2
April	2023	Bitumen	Export	0.79	3.94	0.48
April	2023	Petcoke /	Export	11	9.94	1.22
April	2023	Others%	Export	139.52	734.7	89.44
April	2023	TOTAL PR	Export	4368.52	27417.95	3341.3
April	2023	NET IMPO	Export	18779.12	73792.23	8983.54
May	2023	CRUDE OIL	Import	20705.22	87955.31	10689.25

The dataset captures critical dimensions: time (year, month), country, product category, trade flow (import/export), and volume.

3. Data Processing

The dataset underwent several preprocessing steps:

- **Cleaning:** Removed null values and standardized product names.
- **Filtering:** Separated records into Imports and Exports.
- **Aggregation:** Summed quantities by product year.
- **Derived Metrics:** Calculated total imports and exports per product to determine share percentages.

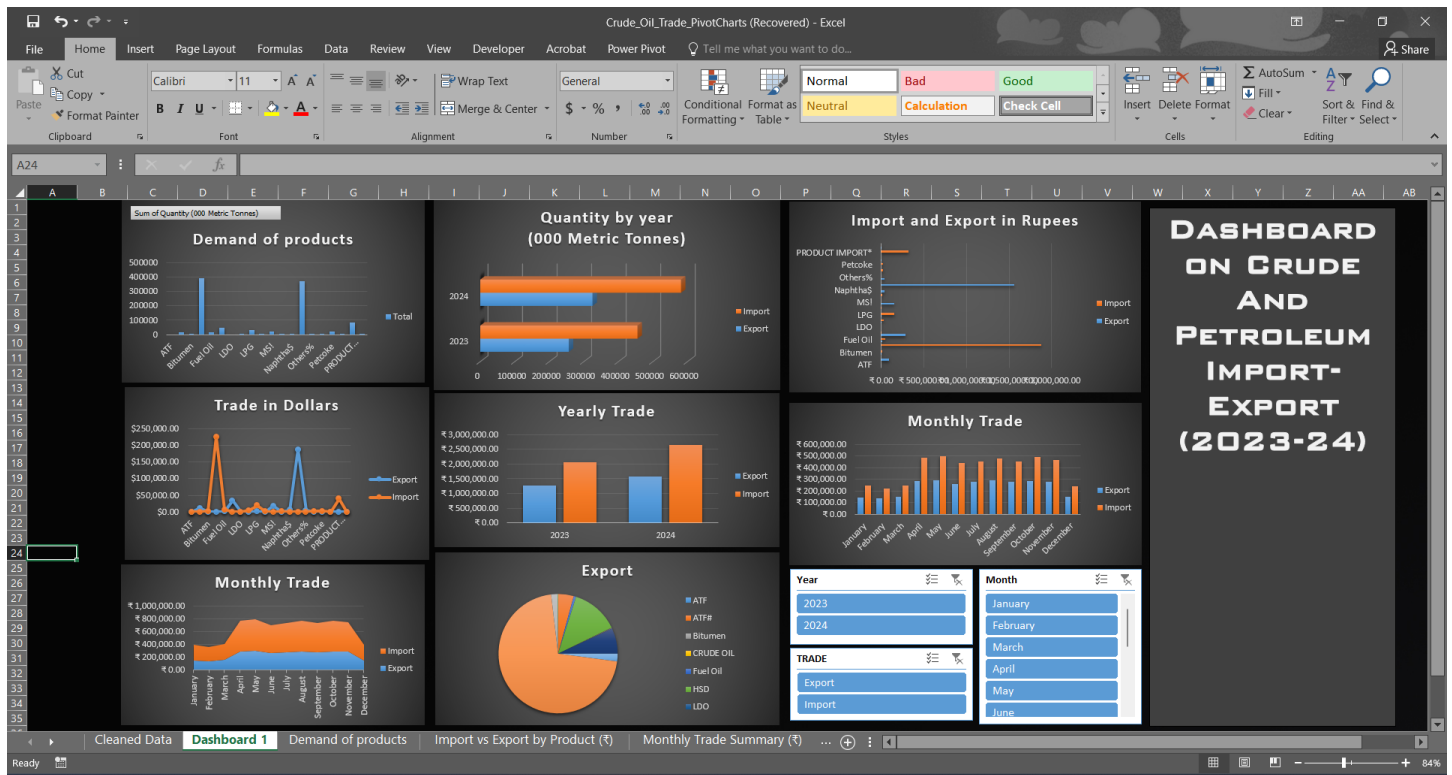
For percentage calculations, we used the formula:

$$\text{Percentage of product} = (\text{Product quantity} / \text{Total quantity for import/export}) \times 100$$

4. Dashboard

Interactive dashboards were created using pivot tables and charts. Key features include:

- Year-wise trends of imports and exports.
- Product-wise contribution in total trade.
- Pie charts displaying percentage shares of each product in total import and export volumes.



A. Data Visualization Descriptions

1. Demand of Products (Bar Chart)

This bar chart visualizes the demand (import and export) across different petroleum products in 000 Metric Tonnes. Crude Oil and Diesel dominate by a wide margin, highlighting India's dependency on crude imports and strength in refining diesel for export. ATF and LPG show moderate activity, while niche products like Bitumen and LDO reflect industrial and regional uses.

2. Quantity by Year (Bar Chart)

This horizontal bar chart compares yearly import and export volumes. In 2023, imports slightly surpassed exports, while in 2024 the gap narrowed—indicating either improved refining output or reduced consumption. The visual suggests a shift toward balancing trade.

3. Import and Export by Product in Rupees (Bar Chart)

Displayed in monetary terms, this chart reveals not just volume but value impact. Imports like Crude Oil and LPG incur high costs, while Diesel, Naphtha, and ATF bring significant export revenue. This split emphasizes refining as a value-adding sector.

4. Trade in Dollars (Line Chart)

A currency-based line chart comparing import and export values in USD across products. Notably volatile, the chart captures international price shifts and currency fluctuations—vital for global trade assessments.

5. Yearly Trade (Column Chart)

A simple yet powerful dual-bar chart showing total export vs import value by year. 2024 shows a higher export total, indicating either better pricing, higher volume, or improved logistics. This could reflect geopolitical shifts or domestic policy changes.

6. Monthly Trade (Stacked Column & Area Charts)

These charts plot trade values month-by-month. Seasonality is evident—peaks around April and October suggest periods of increased industrial or transport demand. The stacked area version gives a cumulative view of import vs export over time.

7. Export Product Breakdown (Pie Chart)

This pie chart illustrates the composition of exports by product. Diesel and Naphtha take the lion's share, showing India's edge in refined fuels. ATF and Fuel Oil trail but are still significant.

8. Slicers (Interactive Filters)

The dashboard includes slicers for Year, Month, and Trade Type. These interactive filters empower users to customize views, spot trends, and dig deeper into product-specific or month-specific trade flows—making the dashboard truly dynamic.

5. Objective wise analysis

A. Product-wise Trade Share

Below is the table showing each product's contribution to total imports and exports:

Product	Import %	Export %
Crude Oil	87.32%	0.00%
Diesel	3.41%	32.58%
Petrol	1.92%	15.79%
Petroleum Naphtha	1.53%	19.77%
Furnace Oil	0.54%	5.98%
ATF (Aviation Turbine Fuel)	0.43%	6.91%
Bitumen	0.48%	3.23%
Others	4.37%	15.74%

B. Time-based Trends

- Crude oil imports peak in Q1 and Q3 of the year.
- Diesel and Naphtha are consistently strong performers in exports throughout the year.

C. Insights and Key Observations

- Crude oil alone accounts for over 87% of imports, underscoring India's dependency on external energy sources.
- Diesel and Petroleum Naphtha lead the export charts, indicating robust refining infrastructure.
- Crude imports peak in Q1 and Q3—potentially linked to seasonal demand and strategic stocking.
- Export destinations are more geographically diverse compared to import sources, which are clustered in the Gulf.
- Aviation fuel exports show modest but steady growth—correlating with the post-pandemic rise in air travel.

6. Conclusion

India's trade in crude oil is dominated by a few key countries in the Gulf region, whereas its petroleum product exports show a more varied geographic spread. Crude oil alone accounts for nearly 87% of import volume, indicating heavy reliance on foreign sources. Export trade is more diversified, with Diesel and Petroleum Naphtha leading the charge.

7. Future Scope and Enhancements

- Integrating pricing data to analyze cost-efficiency and trade margins.
- Forecasting future trade volumes using time-series models.
- Geo-visualization using maps for more intuitive representation.
- Incorporating policy impact analysis (like tariffs or embargoes).

8. References

- Government Trade Records
- Petroleum Planning & Analysis Cell (PPAC)
- Dataset: Crude_Oil_Trade_PivotCharts.xlsx
- Ministry of Petroleum and Natural Gas

Linkedin Link : https://www.linkedin.com/posts/pushkaradityasingh_exceldashboard-datavisualization-petroleum-activity-7319832452032606209-2O1a?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAEgc_r0BJk8ui78Evc1EEU5x37DOybJfINl

Dataset Link: <https://www.data.gov.in/catalog/import-export-petroleum-products>